Group Homework 2:- Shapley Axioms and Fairness

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Contents

0.1 Declaration on the Use of Generative AI

We declare that this was done with integrity and transparency, and that some aspects of GenAI were used as learning aids during brainstorming, preparatory research, and exploratory analysis. Therefore, the final report is a product of our own understanding and synthesis of the concepts, and we have not directly copied any content generated by GenAI. We have cited any sources or references that informed our work, and we have ensured that the report reflects our own critical thinking and insights on the topic.

In this assignment, this response addresses parts (a)-(c) of the assignment using the lecture framing of SHAP as a local, contrastive explanation method and the COMPAS case as a fairness example.

0.2 (a) Efficiency axiom and why $\varphi_{race} = 0$ does not mean the model is race-blind

The **efficiency axiom** says that the Shapley values across all features must add up to the difference between the model's prediction for the instance and the average prediction over the population.

$$\sum_{j=1}^{p} \varphi_j(x) = f(x) - \mathbb{E}[f(X)]$$

This means SHAP explains how the model moved from the *baseline* prediction to the prediction for a particular person. Therefore, $\varphi_{race}(x) = 0$ means that, for this specific case, race receives none of the credit or blame for the gap between $f(x)$ and the average prediction. That is not the same as saying the model is fully race-blind.

So the correct interpretation is: $\varphi_{race}(x) = 0$ if and only if race contributes nothing beyond the average prediction for that instance's allocation. **It does not imply that race never matters anywhere in the system.**

> **Interpretation:** $\varphi_{race}(x) = 0$ means race receives no local credit for the prediction gap in that instance, but does not imply the model is race-blind or fair overall. SHAP is local, can mask proxy effects, and is not proof of fairness.

0.3 (b) Two-feature example where $\varphi_{race} = 0$ but equal FPR parity fails

Let x_1 = race and x_2 = prior arrests. Define a classifier that predicts high risk only from prior arrests:

$$f(\text{race}, \text{prior arrests}) = 1 \text{ if prior arrests} \geq 2, \text{ and } 0 \text{ otherwise.}$$

Because race is never used directly in the prediction rule, race is a dummy feature in the Shapley game. Its marginal contribution is zero for every coalition, so $\varphi_{race}(x) = 0$ for every individual.

Now evaluate **false positive rate (FPR)** only on people with $Y = 0$ (people who do not reoffend):

Group	Prior arrests among $Y = 0$ cases	Predicted positives	FPR
Black	3, 2, 0, 0	2 of 4	0.50
White	1, 0, 0, 0	0 of 4	0.00

Thus, $\text{FPR}_{Black} = 0.50$ while $\text{FPR}_{White} = 0.00$, so equal FPR parity is violated even though race has zero SHAP attribution. The interpretation is important: a model can look race-neutral in a local explanation and still generate racially unequal error burdens because another feature acts as a proxy or is distributed differently across groups.

In the COMPAS context, this helps explain why saying "race was not used" or "race had zero attribution in one case" is not enough. Group-level harm can still occur when a seemingly neutral variable produces systematically higher false positives for one racial group.

> **Key point:** Local SHAP neutrality does not guarantee group-level fairness. Proxy variables or distributional differences can create disparate error rates even when direct attribution is zero. Always check both local and group-level effects.

0.4 (c) Is $\varphi_{race} = 0$ sufficient for fairness?

No. A zero Shapley attribution for race is **not a sufficient condition for fairness**. It only tells us that race did not receive local credit for the prediction relative to the baseline in a particular case, or in some settings across cases. Fairness is broader than local explanation.

- A fair system must also address **proxy discrimination**.
- It must examine **group-level error disparities**, especially in high-stakes settings.
- It should support **contestability, documentation, and monitoring** over time.
- It should distinguish between **transparency** and **substantive fairness outcomes**.

A stronger governance framework would require both: (1) no direct protected-attribute contribution in local explanations, and (2) parity checks on outcomes such as FPR. This is especially appropriate in criminal justice, lending, hiring, insurance, and healthcare, where unequal false positives impose serious social costs.

Under such a framework, I would require both $\varphi_{race} = 0$ and FPR parity as complementary controls. The first guards against direct reliance on race in individual decisions; the second guards against unequal harm at the group level. Together, they form part of a broader model-risk and fairness audit rather than a single definitive fairness test.

Hence, $\varphi_{race} = 0$ and FPR parity are complementary controls. The first prevents direct use of race in individual predictions; the second addresses group-level harm. Both are needed for a robust fairness audit.

0.5 Conclusion

We have shown that the efficiency axiom of SHAP means that $\varphi_{race} = 0$ only indicates no local attribution for that instance, not necessarily fairness. We provided a two-feature example where race has zero SHAP attribution but FPR parity fails, illustrating how proxy variables can create group-level disparities. Finally, we argued that $\varphi_{race} = 0$ is not sufficient for fairness, and that a stronger governance framework would require both local SHAP neutrality and group-level parity checks to ensure a more comprehensive approach to responsible AI in high-stakes domains. Therefore, while SHAP can be a useful tool for local explanations, it should not be the sole criterion for assessing fairness, and must be complemented with group-level analyses and broader governance practices.

0.6 Reference

DNSC 6330 Lecture 02 (course material on SHAP, transparency, and COMPAS).